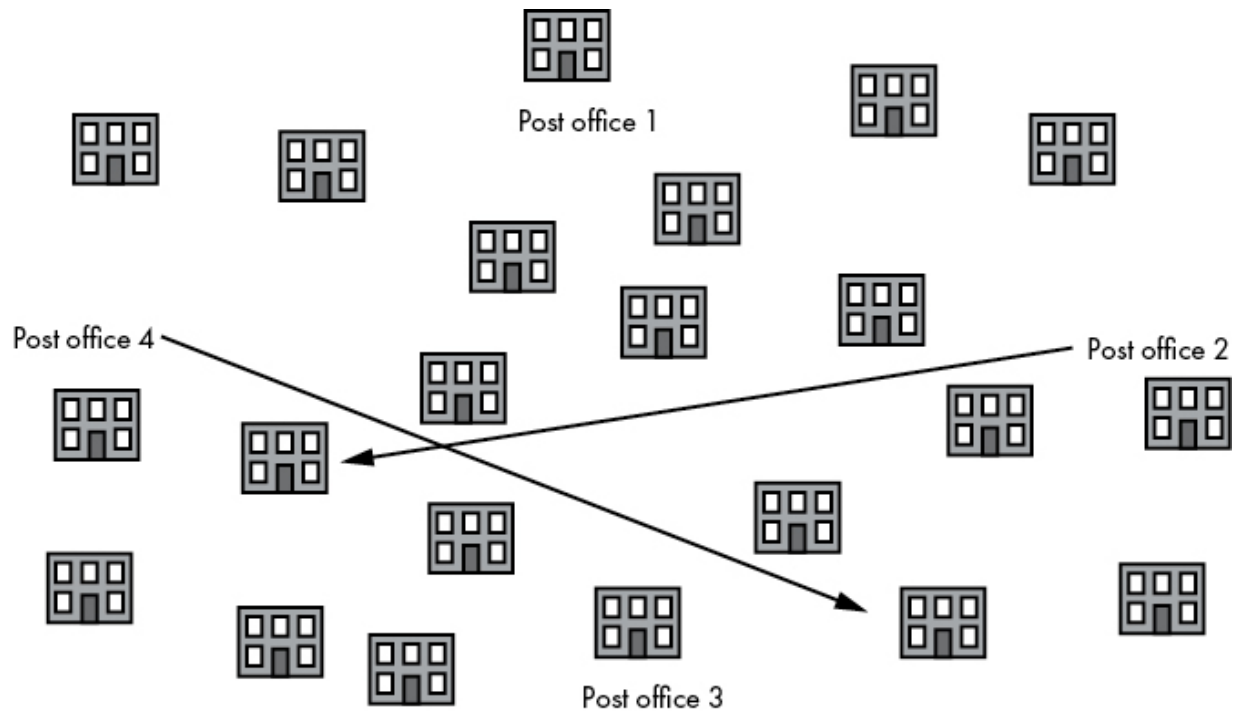




[Figure 7-2:](#) Post offices 2 and 4 have inefficient assignments.

You can rearrange the delivery assignments so that each home receives deliveries from the ideal post office. The ideal post office for a delivery assignment could be the one with the most free staff, the one that possesses suitable equipment for traversing an area, or the one with the institutional knowledge to find all the addresses in an area. But probably, the ideal post office for a delivery assignment is simply the closest one. You may notice that this is similar to the traveling salesman problem (TSP), at least in the sense that we are moving objects around a map and want to decrease the distance we have to travel. However, the TSP is the problem of one traveler optimizing the order of a set route, while here you have the problem of many travelers (letter carriers) optimizing the assignment of many routes. In fact, this problem and the TSP can be solved consecutively for maximum gain: after you make the assignments of which post office should deliver to which homes, the individual letter carriers can use the TSP to decide the order in which to visit those homes.

The simplest approach to this problem, which we might call the *postmaster problem*, is to consider each house in turn, calculating the distance between the house and each of the four post offices, and assigning the closest post office to deliver to the house in question.